

國立屏東大學 114 學年度第二學期 應用物理系 普通物理學(II) 3rd Exam

1. When the geomagnetic intensity is uniformly distributed and its value is 0.5 Gauss, approximately how much magnetic energy is stored in a space 3-m long, 4-m wide, and 2-m high? 1 Tesla = 10^4 Gauss. [10 points]

Magnetic energy density

$$u_B = \frac{B^2}{2\mu_0}$$

$$\Rightarrow \text{Energy} = U_B = u_B V$$

$$= \frac{B^2}{2\mu_0} V$$

Given

$$B = 0.5 \text{ Gauss}$$

and

$$1 \text{ Tesla} = 10^4 \text{ Gauss}$$

$$\therefore B = 0.5 \times 10^{-4} \text{ T}$$

Volume:

$$V = (3)(4)(2) = 24 \text{ m}^3$$

Substituting,

$$U_B = \frac{(0.5 \times 10^{-4})^2}{2(4\pi \times 10^{-7})} (3 \times 4 \times 2)$$

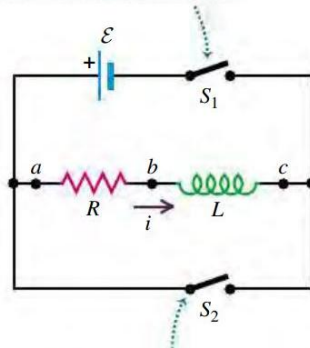
$$= \frac{0.25 \times 10^{-8}}{8\pi \times 10^{-7}} (24)$$

$$= \frac{0.25}{8\pi} \times 10^{-1} (24)$$

$$\approx 0.0239 \text{ J}$$

2. Initially there is no current in the circuit. Switch S_2 is left open, and switch S_1 is closed for a while and the current has reached the value I_0 . Now we close switch S_2 at time $t = 0$, bypassing the battery. The current through R and L does not instantaneously go to zero but decays smoothly. When the current in an LR circuit is decaying, what fraction of the original energy stored in the inductor has been dissipated after 2.3 time constant? [15 points]

Closing switch S_1 connects the R - L combination in series with a source of emf $\mathcal{E}$.



Closing switch S_2 while opening switch S_1 disconnects the combination from the source.

$$\begin{aligned}
 U &= \frac{1}{2} L i^2 = \frac{1}{2} L \left[I_0 e^{-\frac{t}{\tau}} \right]^2 \\
 &= U_0 e^{-\frac{2t}{\tau}} \\
 &= U_0 e^{-\frac{2}{\tau}(2.3\tau)} \\
 &= U_0 e^{-4.6} = 0.01 U_0 \rightarrow \text{energy dissipated is } 99\%
 \end{aligned}$$

3. The length of a 50 mH cylindrical inductor in the laboratory is about 1.2 cm and the radius is about 0.35 cm. If it is regarded as an ideal solenoid and the internal medium of the solenoid is air, please estimate the number of turns of this inductor. [10 points]

Self-inductance of a solenoid

$$L = \frac{\mu_0 N^2 A}{l}$$

$$\Rightarrow N = \sqrt{\frac{Ll}{\mu_0 A}}$$

Given:

$$L = 50 \text{ mH} = 50 \times 10^{-3} \text{ H}$$

$$l = 1.2 \text{ cm} = 1.2 \times 10^{-2} \text{ m}$$

$$r = 0.35 \text{ cm} = 0.35 \times 10^{-2} \text{ m}$$

$$A = \pi r^2 = \pi (0.35 \times 10^{-2})^2$$

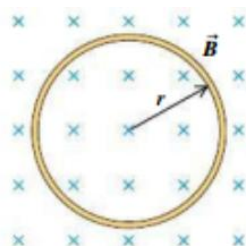
So,

$$N = \sqrt{\frac{(50 \times 10^{-3})(1.2 \times 10^{-2})}{(4\pi \times 10^{-7})\pi(0.35 \times 10^{-2})^2}}$$

$$N \approx 3.5 \times 10^3$$

$$N \approx 3520 \text{ turns}$$

4. A circular loop of wire with radius $r = 0.025 \text{ m}$ and resistance $R = 0.39 \Omega$ is in a region of spatially uniform magnetic field. The magnetic field is directed into the plane of the figure. At $t = 0$, $B = 0$. The magnetic field then begins increasing, with $B(t) = 0.380t^3 \text{ T}$, where t is measured in seconds. What is the current in the loop (magnitude and direction) at the instant when $B = 1.33 \text{ T}$? [10 points]



IDENTIFY: The changing flux through the loop due to the changing magnetic field induces a current in the wire.

SET UP: The magnitude of the induced emf is $|\mathcal{E}| = \left| \frac{d\Phi_B}{dt} \right| = \pi r^2 \left| \frac{dB}{dt} \right|$, $I = \mathcal{E}/R$.

EXECUTE: $\vec{B}$ is into the page and Φ_B is increasing, so the field of the induced current is directed out of the page inside the loop and the induced current is counterclockwise.

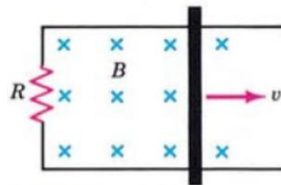
$$|\mathcal{E}| = \left| \frac{d\Phi_B}{dt} \right| = \pi r^2 \left| \frac{dB}{dt} \right| = \pi (0.0250 \text{ m})^2 (0.380 \text{ T/s}^3) (3t^2) = (2.238 \times 10^{-3} \text{ V/s}^2) t^2.$$

$$I = \frac{|\mathcal{E}|}{R} = (5.739 \times 10^{-3} \text{ A/s}^2) t^2. \text{ When } B = 1.33 \text{ T, we have } 1.33 \text{ T} = (0.380 \text{ T/s}^3) t^3, \text{ which gives}$$

$$t = 1.518 \text{ s. At this } t, I = (5.739 \times 10^{-3} \text{ A/s}^2) (1.518 \text{ s})^2 = 0.0132 \text{ A.}$$

EVALUATE: As the field changes, the current will also change.

5. A metal rod of mass m and length L slides on a frictionless rail of negligible resistance, with a resistor R at the termination point (see figure). A uniform magnetic field B is directed perpendicular to the plane of the rails. The initial velocity of the rod is v_0 . There is no external force on this rod. Find the velocity of the rod as function of time. [10 points]



$$\begin{aligned} F_B &= m\ddot{x} = m\dot{v} \\ \Rightarrow -\frac{B^2 L^2}{R} v &= m \frac{dv}{dt} \\ \Rightarrow -\frac{B^2 L^2}{mR} dt &= \frac{1}{v} dv \\ \Rightarrow -\frac{B^2 L^2}{mR} \int_0^t dt &= \int_{v_0}^v \frac{1}{v} dv \\ \Rightarrow -\frac{B^2 L^2}{mR} t &= \ln\left(\frac{v}{v_0}\right) \Rightarrow v(t) = v_0 e^{-\frac{B^2 L^2}{mR} t} \end{aligned}$$

6. The maximum radiation power of a mobile phone is 0.8 W. If the minimum peak electric field that the base station receiver can handle is 1.2 mV/m, what is the maximum allowable distance between the mobile phone and the base station? [10 points]

Using

$$S_{\text{avg}} = \frac{1}{2} \epsilon_0 c E_0^2$$

and

$$P = S_{\text{avg}}(4\pi R^2)$$

$$P = \frac{1}{2} \epsilon_0 c E_0^2 (4\pi R^2)$$

$$\Rightarrow R = \sqrt{\frac{P}{2\pi \epsilon_0 c E_0^2}}$$

Substitute:

$$P = 0.8$$

$$E_0 = 1.2 \times 10^{-3}$$

$$\epsilon_0 = 8.85 \times 10^{-12}$$

$$c = 3.0 \times 10^8$$

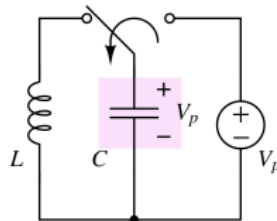
$$R = \sqrt{\frac{0.8}{2\pi(8.85 \times 10^{-12})(3.0 \times 10^8)(1.2 \times 10^{-3})^2}}$$

$$R \approx \sqrt{3.32 \times 10^7}$$

$$R \approx 5.76 \times 10^3 \text{ m}$$

$$R_{\text{max}} \approx 5.8 \text{ km}$$

7. A 24 μF capacitor is charged by a $V_P = 170 \text{ V}$ power supply, then disconnected from the power and connected in series with a 0.21 mH inductor. Calculate: (a) the oscillation frequency of the circuit; (b) the energy stored in the capacitor at time $t = 0$ (the moment of connection with the inductor); (c) the energy stored in the inductor at $t = 1.35 \text{ ms}$. [15 points]



IDENTIFY: An L - C circuit oscillates, with the energy going back and forth between the inductor and capacitor.

(a) **SET UP:** The frequency is $f = \frac{\omega}{2\pi}$ and $\omega = \frac{1}{\sqrt{LC}}$, giving $f = \frac{1}{2\pi\sqrt{LC}}$.

EXECUTE: $f = \frac{1}{2\pi\sqrt{(0.210 \times 10^{-3} \text{ H})(24.0 \times 10^{-6} \text{ F})}} = 2.242 \times 10^3 \text{ Hz}$, which rounds to 2.24 kHz.

(b) **SET UP:** The energy stored in a capacitor is $U = \frac{1}{2} CV^2$.

EXECUTE: $U = \frac{1}{2}(24.0 \times 10^{-6} \text{ F})(170.0 \text{ V})^2 = 0.347 \text{ J}$.

(c) **SET UP:** The current in the circuit is $i = -\omega Q \sin \omega t$, and the energy stored in the inductor is $U = \frac{1}{2} Li^2$.

EXECUTE: First find ω and Q . $\omega = 2\pi f = 2\pi(2242 \text{ Hz}) = 1.409 \times 10^4 \text{ rad/s}$.

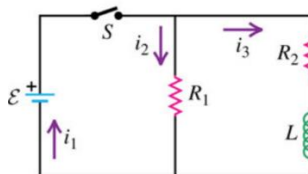
$Q = CV = (24.0 \times 10^{-6} \text{ F})(170.0 \text{ V}) = 4.08 \times 10^{-3} \text{ C}$. Now calculate the current:

$i = -(1.409 \times 10^4 \text{ rad/s})(4.08 \times 10^{-3} \text{ C})\sin[(1.409 \times 10^4 \text{ rad/s})(1.35 \times 10^{-3} \text{ s})]$. Notice that the argument of the sine is in *radians*, so convert it to degrees if necessary. The result is $i = -9.84 \text{ A}$.

Now find the energy in the inductor: $U = \frac{1}{2} Li^2 = \frac{1}{2}(0.210 \times 10^{-3} \text{ H})(-9.84 \text{ A})^2 = 0.010 \text{ J}$.

EVALUATE: At the end of 1.35 ms, more of the energy is now in the inductor than in the capacitor.

8. An inductor with inductance $L = 0.3 \text{ H}$ and negligible resistance is connected to a battery, a switch S , and two resistors, $R_1 = 12 \Omega$ and $R_2 = 16 \Omega$. The battery has emf 96 V and negligible internal resistance. S is closed at $t = 0$. (a) What are the currents i_1 , i_2 and i_3 just after S is closed? (b) What are i_1 , i_2 and i_3 after S has been closed a long time? (c) What is the value of t for which i_3 has half of the final value that you calculated in part (b)? (d) When i_3 has half of its final value, what are i_1 and i_2 ? [20 points]



IDENTIFY and SET UP: Apply $\mathcal{E}_L = -L \frac{di}{dt}$ and $V_R = Ri$. An inductor opposes a change in current through it. Kirchhoff's rules apply.

EXECUTE: (a) At the instant the switch is closed, the inductor will not allow any current through it, so all the current goes through R_1 . So $i_1 = i_2 = \mathcal{E}/R_1 = (96.0 \text{ V})/(12.0 \Omega) = 8.00 \text{ A}$. $i_3 = 0$.

(b) After a long time, steady-state is reached, so $di_3/dt = 0$ and $\mathcal{E}_L = -L \frac{di_3}{dt} = 0$. In this case, the potential across R_1 and across R_2 is 96.0 V . Therefore

$$i_2 = (96.0 \text{ V})/(12.0 \Omega) = 8.00 \text{ A}.$$

$$i_3 = (96.0 \text{ V})/(16.0 \Omega) = 6.00 \text{ A}.$$

$$i_1 = i_2 + i_3 = 8.00 \text{ A} + 6.00 \text{ A} = 14.00 \text{ A}.$$

(c) Apply Kirchhoff's loop rule, giving

$$\mathcal{E} - i_3 R_2 - L \frac{di_3}{dt} = 0.$$

Separating variables and integrating gives

$$\int_0^t -\frac{R_2}{L} dt' = \int_0^{i_3} \frac{1}{i_3' - \mathcal{E}/R_2} di_3'.$$

Carrying out the integration and solving for t gives

$$-\frac{R_2}{L} t = \ln \left(\frac{i_3 - \mathcal{E}/R_2}{-\mathcal{E}/R_2} \right).$$

$$t = \frac{L}{R_2} \ln \left(\frac{\mathcal{E}/R_2}{\mathcal{E}/R_2 - i_3} \right) = \frac{0.300 \text{ H}}{16.0 \Omega} \ln \left(\frac{96.0 \text{ V}}{96.0 \text{ V} - (3.00 \text{ A})(16.0 \Omega)} \right) = 0.0130 \text{ s} = 13.0 \text{ ms}.$$

(d) $i_2 = \mathcal{E}/R_1 = (96.0 \text{ V})/(12.0 \Omega) = 8.00 \text{ A}$. $i_1 = i_2 + i_3 = 8.00 \text{ A} + 3.00 \text{ A} = 11.0 \text{ A}$.

EVALUATE: At steady-state, the potential drop across an inductor is zero if it has no resistance. Initially the inductor acts like an open circuit because it will not allow current to flow through it.